

# Note on ROS2 Installation on Windows

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## Installation Steps

In Powershell, do

```
irm https://pixi.sh/install.ps1 | iex
```

Once installed, restart Powershell

create ros environment in Powershell

```
pixi init roswin  
cd roswin
```

Go to the C Drive -> User -> YourName -> roswin folder, open `pixi.toml` using text editor

Paste this into

```
[workspace]  
name = "roswin"  
channels = ["conda-forge", "robostack-staging"]  
platforms = ["win-64"]  
  
[dependencies]  
ros-humble-desktop = "*"   
ros-humble-moveit = "*"
```

Then in Powershell,

```
pixi install
```

Close Powershell, then reopen. Remove safety check

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

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## Verify ROS Installation

☐ Check Basic ROS

```
cd C:\Users\YourName\roswin  
pixi shell
```

```
ros2 --help
```

You will see a list help commands.

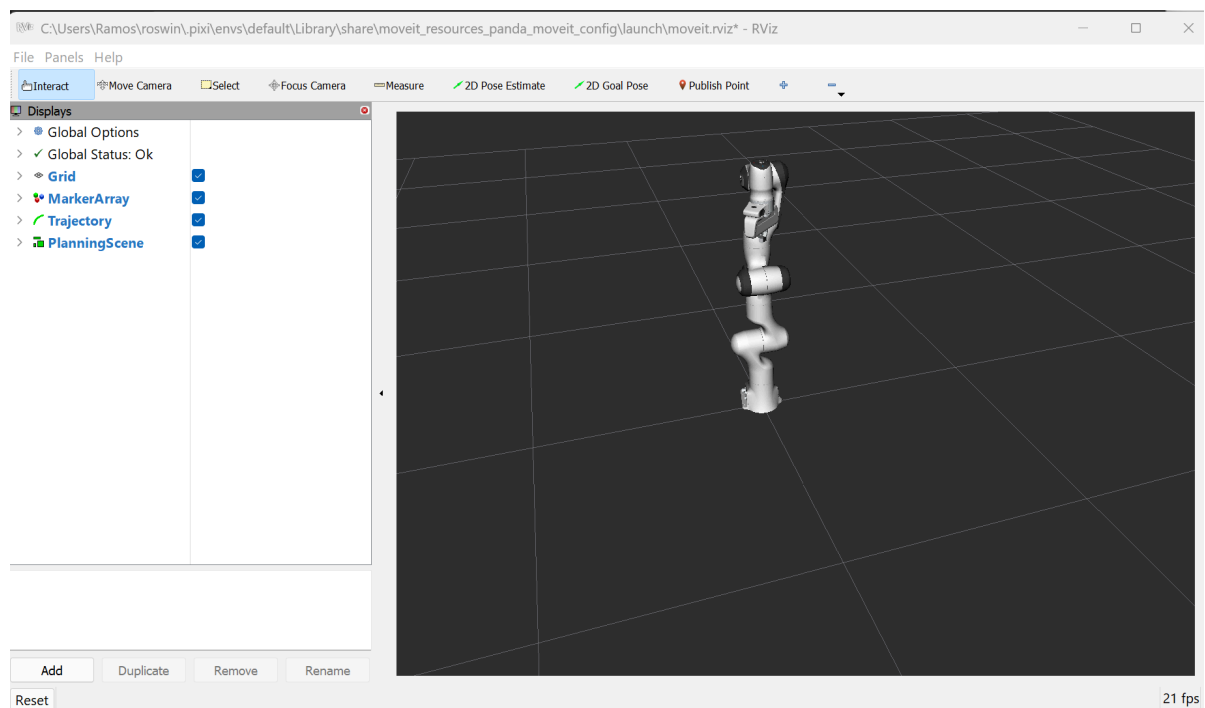
☐ Check MoveIt GUI

In the pixi shell,

```
cd C:\Users\YourName\roswin  
pixi shell
```

```
pixi add ros-humble-moveit-resources-panda-moveit-config  
pixi install  
ros2 launch moveit_resources_panda_moveit_config demo.launch.py
```

You will see this GUI.



## Serial Communication Check

Install python serial

```
python -m ensurepip --upgrade  
python -m pip install pyserial
```

Get an Arduino and connect with PC via USB

For Arduino, do this code:

```
void setup() {  
  Serial.begin(115200);  
}  
  
void loop() {  
  Serial.println("ARDUINO_OK");  
  delay(500);  
}
```

In pixi shell, do

```
python -c "import serial,time,os; s=serial.Serial('COM7',115200,timeout=1);  
log='log.csv'; t0=time.time(); print('Logging... Ctrl+C to stop');  
while time.time()-t0<5:  
  r=s.readline().decode(errors='ignore').strip();  
  if r: open(log,'a').write(f'{time.time()},{r}\n'); print(r)"
```

You should see messages from Arduino "ARDUINO\_OK".